Language Model-Based Framework for Automated Data Extraction and Quality Control in Oncological Cancer Registries

Ph.D. Project Proposal

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1 State of the art

This section provides an overview of the current state of the art in AI applications for oncology. We start by explaining traditional AI technologies such as Machine Learning, and then discuss the emergence of Large Language Models (LLMs) and their current applications in oncology.

Machine Learning Machine Learning is a subfield of Artificial Intelligence. It focuses on creating systems that learn from data and improve their performance over time without being explicitly programmed for a specific task.

Machine Learning gained widespread popularity during the 1990s and early 2000s with the introduction of popular research such as SVM [1] and Random Forests [2] method. Machine Learning has proven effective across a wide range of tasks, finding applications in classification, clustering, regression and anomaly detection, which align well with real-world problems in many different fields such as medical syndrome detection, traffic sign recognition in autonomous driving systems, financial fraud detection.

The popularity of this technology led to the introduction of neural network [3]. It is a new model of Machine Learning and it is inspired by the structure of the human brain.

Neural networks set a new milestone for Machine Learning because they expanded the scope of its applications to high-dimensional data such as images, videos and complex structured data.

Large Language Models Large Language Models (LLMs) are a subcategory of Artificial Intelligence (AI), and more specifically, they belong to the field of Deep Learning [4].

In recent years, they have become extremely popular thanks to their ability to understand and generate natural language with a high level of fluency and coherence.

Their generalized knowledge and versatility make them perfect for a wide range of applications, from text generation and translation to code assistance, question answering, and conversational systems. The ability to generate natural language aligns well with the need to create systems that can be used by everyone without requiring specific technical knowledge.

Unlike traditional software interfaces, which often require predefined commands or structured inputs, LLMs allow users to interact using natural language, making human-computer interaction more intuitive and accessible.

An extension of classic LLMs is represented by Multimodal Models [5]. While LLMs only generate text from a given text input, Multimodal Models are capable of processing not only text, but also images, audio and other data types within the same model. This feature extends their application to a wider range of use cases and opens new possibilities for integrating AI diagnostic tools into the medical workflow.

Recent evolutions of LLMs include the introduction of AI Agents. Their ability to autonomously plan and execute multi-step tasks enables them to perform complex workflows with minimal human interaction. AI Agents can also interact with external tools such as APIs, databases and other external resources, further expanding the capabilities of LLMs [6].

Around the same period, another important AI technology emerged, the Retrieval Augmented Generation (RAG) [7]. RAG introduced a new way to enhance the response of Generative AI. In particular, when a query is processed, the system first retrieves relevant documents or knowledge snippets from an external verified database (such as medical guidelines, clinical repositories, or local patient records). This novelty enables the LLM to generate fact-based responses [7].

Today's oncology research In recent years, oncology research has increasingly adopted Machine Learning and LLMs as integrated tools for cancer diagnosis. In particular, studies show that Machine Learning models have high performance sometimes approaching 100% accuracy in cancer detection [8], with the most studied cancers being breast cancer, lung cancer, followed by brain, cervical and liver cancer, using mainly CNN as the primary architecture [8].

A systematic literature review of February 2026 shows that ML approaches remain concentrated on well-studied cancer types such as breast and lung cancer, while other rare or low reproducibility cancers remain underrepresented or understudied in the literature [9].

While ML classifiers achieved reasonable success in specific tasks such as thyroid carcinoma diagnosis, their dependency on manual feature extraction limited their generalizability and robustness against data variability, creating a bottleneck in this research field. This gap led to the research in Large Language Models. Research on LLMs in oncology has increased significantly in recent years, due to their wide applicability such as the ability to process unstructured data, their generalization capabilities and their ability to generate natural language, enabling automated extraction of relevant information from clinical texts and supporting clinical decision-making processes [10]. This technology could also resolve a common problem of ML-based diagnostic tools, which is the lack of explainability and then, the trust of such tools by medical professionals [10]. However, studies show several critical issue regarding the use of LLM in this field. The main concern is hallucination that may produce during the generation of the responses [10].

These technologies can be both integrated in the medical worflow for automating the data extraction (LLM), identifying possible anomalies (ML/LLM) and supporting decision-making processes (LLM).

1.1 Previous experience

During my Master's degree I became curious about AI and its possible applications while attending the "Machine Learning" course. This led me to follow the "Intelligent Systems Engineering" course and eventually to write my Thesis on Large Language Model Architectures.

My passion for sports and medicine led me to think about possible applications of AI in the medical field.

2 Project Description

2.1 Motivations and Problem Statement

As outlined in the section 1, Artificial Intelligence has shown promising results in oncology. Machine Learning and Large Language Models have demonstrated their potential in cancer detection, information extraction from clinical documents, and clinical decision support. Despite these advances, several challenges still limit the effective adoption of AI technologies in oncology research and clinical practice.

First, a large portion of oncology information is stored in unstructured clinical documents and pathology reports. The extraction of information from these documents is still largely performed manually. This issue affects oncology research because the manual extraction of data is time-consuming and may introduce human errors in the process, creating a bottleneck for later stages of the research process.

Second, the quality and consistency of oncology datasets remain critical issues. Inaccurate or incomplete data can negatively affect both clinical activities and the development of AI models for diagnostic support.

Finally, a recent systematic review provides evidence that LLMs can improve clinical workflows and support healthcare professionals in several tasks. However, despite these promising results, their adoption in real-world clinical settings remains limited[11].

2.2 Idea

The idea of this project aim to fill the gap by developing technologies that can be integrated in the medical workflow, with a particular focus on oncology. The core of the project is the design and implementation of a framework architecture that includes:

Automation of Data Extraction The first important component of the architecture is the automation of data extraction from clinical documents. This component will be based on a Language Model that will be used to extract relevant information. Its usage will be focused on the extraction, but also on the structuring of such data, aiming to organize this information in widely adopted standard formats such as ICD-O-3[1] and TNM[2].

Data Quality and Discrepancies Detection The second component of the architecture includes the identification of possible human errors in the data. This component will ensure the quality of the structured data and it will be based on a Language Model that will be used to identify anomalies in already existing data. The use of LLMs/SLMs improve the speed of identification process and it generates natural language explanations of the possible errors, facilitating the human-in-the-loop approach for the supervision of the process.

Diagnostic Support If the previous components are successfully implemented and evaluated in a modest time, the architecture will be extended to include a diagnostic support component. If the architecture has a good performance in ensuring data quality, it would also address one of the main bottlenecks in ML-based cancer diagnosis: the lack of reliable and structured training data. This would lay the foundation for training a Machine Learning model for diagnostic support, which would then be integrated as an extension of the framework.

Integration of the architecture in a real medical workflow A core focus of this project is ensuring high potential for real-world clinical adoption. In order to ensure that, standard formats will be used and the framework must remain generic and extendible. In case of collaborations with public istitutes, the architecture will be adapted in order to maximize the usability for specific requests.

[1] `https://www.who.int/standards/classifications/other-classifications/international-classification-of-diseases-for-oncology`
[2] `https://www.uicc.org/resources/tnm`

This architecture is designed to be a tool for medical professionals. For this reason, it is designed to automate a manual process, with the aim of improving the efficiency and the quality of the data, but without replacing the human supervision.

2.3 Challenges

Architecture weaknesses This architecture may face various challenges. As mentioned in the section 1 hallucination is a well-known problem in medical LLM applications. It is particularly relevant in this context because it could lead to a deterioration of output quality and a consequent decrease in trustworthiness. A mitigation strategy will include the use of a RAG approach.

The framework may also encounter incomplete or ambiguous clinical documents, which could reduce the quality of the output. A robust architecture should handle these situations by identifying such cases and signaling the need for human supervision, avoiding the automatic gap-filling that LLMs tend to perform.

LLM Privacy and Security The use of LLMs in healthcare involves significant privacy and security concerns, in particular, regarding the European GDPR regulation. For this reason, a strategy for the use of LLMs will be defined. A possible strategy is to use open-source Small Language Models (SLMs) such as LLaMa[3] or Mistral[4], that can be trained and used locally without the need of sending data to external servers or enterprises. These SLMs are nowadays very powerful and they can be used for many different tasks. LLMs can also be used for different tasks that do not involve the use of sensitive data, such as data synthesis.

These privacy and security concerns are also strictly related to the specific compliance constraints of future collaborators and the hardware possibilities of the research group or the collaborators. A strong hardware could permit the use of local Large or Medium Language Models, providing better performances and a wider range of applications, while a weaker hardware consequently limits the use of SLMs with a possible decrease of the performance.

Data Retrieval One of the main challenges of medical research is how the data is retrieved and manipulated. In particular, the data should represent real cases and be of high quality, but at the same time it should be anonymized to protect the privacy of patients.

A well known strategy to overcome the data retrieval problem is to collaborate with real public healthcare institutions. Such collaborations require time and includes the

[3] `https://www.llama.com/`
[4] `https://mistral.ai/`

process of access and anonymization of the data that should be taken into consideration. For this reason, part of the project will be dedicated to the activity of searching for collaborations with these institutes such as Ausl Romagna and IRST of Meldola and the anonymization process.

A fallback strategy will be also defined in case the collaborations are not found in time. This strategy includes the use of publicly available datasets such as TCGA dataset[5], NCI Registry[6] or Belgium Cancer Registry[7] (subject to a formal request) and it could also include the generation of synthetic data and the verification of its realism.

3 Expected Results

Technical Deliverables On a practical level, the expected results of this project include:

1. The development of a framework architecture for oncology purposes that automates the data extraction process, enhancing the quality of the data, identifying possible human errors.
2. Adapting such framework in order to integrate it in a real medical workflow.
3. Evaluation of the developed framework against real-world cases, identifying the scenarios in which the automated pipeline performs best and those where human supervision remains essential.

Scientific Contributions During the first year, a systematic literature review in the field of LLM and ML application on the oncology field will be created and published. This will help to increase the personal knowledge for a complete understanding of the state of the art and to identify the gap that will be filled by the project. Then, when the architecture is complete, a scientific paper describing the architecture and its application will be published. Finally, the evaluation will be conducted and the results will be published in another scientific paper.

4 Subdivision of the project and timeline

The project idea is divided into three main phases, corresponding to the three available years:

1. **Knowledge acquisition and collaboration search** (first year)
   - The study of literature will be consistent and it will produce the systematic literature review described in the previous section.

[5] `https://github.com/jkefeli/tcga-path-reports`
[6] `https://portal.gdc.cancer.gov/`
[7] `https://kankerregister.org/en`

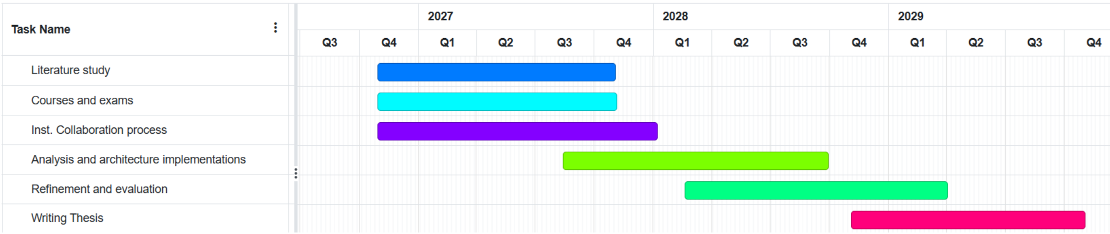

Figure 1: Project timeline

   - Phd courses will be chosen and followed in order to acquire more knowledge passing the exams.
   - The search of collaborations will be conducted in parallel with the knowledge acquisition and it continues with the process of data anonymization.

2. **Design and implementation of the framework** (second year)
   - Design and implementation of the proposed framework.
   - Refinement of the framework and its adaptation with the collaborators feedbacks for the integration in the medical workflow.
3. **Evaluation and dissemination** (third year)
   - Evaluation and dissemination of the developed framework against real-world cases.
   - Write final thesis.

5 Proposed evaluation criteria

The proposed evaluation criteria include three main categories of criteria:

- **Scientific criteria**: the project is successful if at least the scientific contribution explained in section 3 are achieved.
- **Performance criteria**: the framework will be evaluated using performance criteria for each part of the process, including the evaluation of the quality of data extraction and structured output, percentage of discrepancies detection out of the total and the accuracy of diagnostic support output. These metrics are considered the main ones, but each of them has sub metrics that includes the generation of false negatives and false positives.
- **Real scenario criteria**: if the architecture is deployed in a real scenario, a positive evaluation include a significant reduction in the time spent by medical professionals on specific stasks and if the framework has a concrete impact on the efficiency of the collaborators.

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